

WEEK 1 TASK

DATA COLLECTION AND CLEANING STRATEGY

Junior Data Analyst – Agribusiness Virtual Internship

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Internship: Junior Data Analyst – Agribusiness Virtual Intern

Week: 1

Task: Data Collection and Cleaning Strategy

1. Introduction

Agribusiness generates large amounts of data related to crop production, agricultural prices, weather conditions, land use, yield, supply chains, and market activities. Proper collection and cleaning of this information is important for producing reliable analysis and supporting data-driven decision-making.

This report presents a proposed data collection and cleaning strategy for agribusiness-related data. The strategy focuses on identifying reliable public data sources, selecting appropriate collection methods, maintaining data quality, and preparing datasets for further analysis.

The report is designed as a planning framework for subsequent data analysis activities. No claim is made that the datasets discussed in this report have already been collected or cleaned.

2. Task Objectives

The main objectives of this task are:

- Identify relevant publicly available sources of agribusiness data.
 - Develop a systematic approach for collecting agricultural data.
 - Describe suitable methods such as direct downloads, APIs, and web scraping.
 - Develop a structured data-cleaning strategy.
 - Identify common data-quality problems.
 - Define methods for handling missing values, duplicates, and outliers.
 - Compare manual and automated data-cleaning approaches.
 - Identify appropriate tools and technologies.
 - Establish a data-quality validation framework.
 - Prepare a reliable foundation for future exploratory and statistical analysis.
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3. Proposed Agribusiness Data Topic

For this internship, the proposed analytical area is:

Agricultural Crop Production, Yield and Price Data

The analysis can focus on information such as:

- Crop type
- Crop production
- Crop yield
- Cultivated area
- Agricultural year
- State or region
- Market price
- Minimum and maximum price
- Weather-related variables where available
- Commodity or crop category

This topic provides opportunities for analyzing production trends, regional differences, price movements, relationships between variables, and future trends.

4. Data Requirements

The proposed dataset should ideally contain the following categories of information:

Category	Example Variables
Crop Information	Crop name, crop category
Location	State, district, region
Production	Production quantity
Yield	Yield per hectare
Area	Cultivated area
Time	Year, month, agricultural season
Price	Minimum price, maximum price, average price
Weather	Rainfall, temperature, if available
Market Information	Market/location, commodity

The exact variables will depend on the selected public dataset.

5. Potential Public Data Sources

At least three publicly available sources should be considered before selecting the final dataset.

5.1 Government Open Data Sources

Government open-data portals can provide datasets related to agriculture, crops, production, markets, and other public-sector information.

One important source for India is the Open Government Data Platform (data.gov.in).

Potential uses include:

- Agricultural production data
- Crop-related statistics
- State and district-level information
- Government reports and datasets

The final dataset should be evaluated for completeness, update frequency, metadata, and suitability for the selected analysis.

5.2 Directorate of Economics and Statistics / Agricultural Statistics

Government agricultural statistics and publications can provide information about:

- Crop production
- Area under cultivation
- Yield
- Agricultural trends
- State-level statistics

These sources can be useful for validating information obtained from other datasets.

5.3 Agricultural Market and Commodity Data

Agricultural market information can provide information about commodity prices and market activity.

Potential variables include:

- Commodity
- Market
- Location
- Date

- Minimum price
- Maximum price
- Modal or average price

Such data can support analysis of agricultural price fluctuations and market trends.

5.4 Weather and Climate Data

Weather data can be considered as a supporting data source.

Potential variables include:

- Rainfall
- Temperature
- Humidity
- Weather conditions
- Date
- Location

Weather information can potentially be combined with agricultural data to study relationships between environmental conditions and agricultural outcomes.

6. Source Selection Criteria

Before using a dataset, the following criteria should be considered:

6.1 Reliability

The source should be an established government, institutional, academic, or otherwise credible public data provider.

6.2 Relevance

The dataset should contain variables directly related to the selected agribusiness question.

6.3 Completeness

Important variables should have sufficient observations and limited missing information.

6.4 Timeliness

The dataset should contain reasonably recent information when the analysis requires current or recent trends.

6.5 Consistency

Variables should follow consistent naming, units, formats, and definitions.

6.6 Accessibility

The data should be legally accessible through downloads, APIs, or other permitted methods.

7. Data Collection Methods

Three primary collection methods are considered.

7.1 Direct Download

If a reliable source provides CSV, Excel, JSON, or other downloadable files, direct downloading is the preferred method.

Process

1. Identify the relevant dataset.
2. Review its description and metadata.
3. Check the publication date and update frequency.
4. Download the dataset.
5. Record the source and download date.
6. Store the original file without modification.
7. Create a working copy for analysis.

Advantages

- Simple to implement
- Easy to reproduce

- Usually requires less technical infrastructure
- Suitable for static datasets

Potential Challenges

- Dataset may not be regularly updated.
 - File formats may differ.
 - Data definitions may change between versions.
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8. API-Based Data Collection

An API can be used when the data provider makes structured programmatic access available.

Proposed Process

1. Identify an appropriate API.
2. Read the API documentation.
3. Identify required endpoints and parameters.
4. Request the required data.
5. Check the response format.
6. Convert the response into a structured dataset.
7. Store the retrieved data with source and retrieval-date information.
8. Validate the returned records.

Advantages

- Can automate data collection.
- Useful for regularly updated datasets.
- Reduces repetitive manual downloading.
- Can retrieve only the required information.

Potential Challenges

- API limits
 - Authentication requirements
 - Changes to API structure
 - Network failures
 - Incomplete responses
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9. Web Scraping

Web scraping may be considered when relevant public information is available on websites but no suitable download or API is provided.

Proposed Process

1. Identify the relevant webpage.
2. Check whether automated collection is permitted.
3. Understand the webpage structure.
4. Extract required information.
5. Convert extracted information into a structured format.
6. Clean and validate the collected data.
7. Store the source URL and collection date.

Important Considerations

Web scraping should respect:

- Website terms of use
- Robots.txt where applicable
- Access restrictions
- Copyright and licensing
- Rate limits
- Privacy requirements

Web scraping should not be the first choice when an official downloadable dataset or API is available.

10. Data Collection Workflow

The proposed data collection workflow is:

Define Business Question

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Identify Required Variables

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Find Reliable Public Sources

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Evaluate Source Quality

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Select Collection Method

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Collect Data

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Record Source and Metadata

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Store Original Dataset

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Create Working Dataset

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Begin Data Cleaning

11. Data Collection Challenges and Solutions

Challenge	Potential Solution
Different data formats	Convert files into a common structured format
Missing metadata	Review documentation or source descriptions
Different measurement units	Standardize units before analysis
Duplicate records	Identify duplicate keys and remove verified duplicates
Inconsistent naming	Create standardized categories
Outdated information	Record publication/update dates
API limits	Use controlled requests and appropriate batching
Website structure changes	Validate scraping scripts regularly
Missing values	Identify patterns before deciding treatment
Multiple data sources	Compare definitions and standardize variables

12. Data Cleaning Strategy

Data cleaning is the process of identifying and correcting problems that could affect the accuracy of analysis.

The proposed cleaning process consists of the following stages:

1. Data inspection
2. Structural validation
3. Missing-value analysis
4. Duplicate detection
5. Data-type correction
6. Standardization

7. Outlier detection
 8. Logical validation
 9. Final quality checks
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13. Initial Data Inspection

Before modifying the dataset, the following checks should be performed:

- Number of rows
- Number of columns
- Column names
- Data types
- Unique values
- Missing values
- Duplicate records
- Minimum and maximum values
- Basic descriptive statistics

This initial inspection helps identify potential problems before changes are made.

14. Handling Missing Values

Missing values can occur because information was not recorded, was unavailable, or was lost during data collection.

Proposed process

First, calculate the percentage of missing values for each variable.

Missing Value Level	Proposed Action
Very low	Consider appropriate imputation
Moderate	Investigate the reason for missingness
High	Consider excluding the variable if justified

Critical variable with high missingness Investigate source or alternative dataset

Possible Techniques

For numerical variables:

- Mean imputation
- Median imputation
- Group-based imputation

For categorical variables:

- Mode
- “Unknown” category

For time-series data:

- Forward filling or interpolation may be considered where appropriate.

The selected method should depend on the nature of the variable and the reason for missingness.

15. Duplicate Data Handling

Duplicate records can incorrectly increase the importance of observations.

Proposed process

1. Identify completely duplicated rows.
2. Identify potential duplicates using key fields.
3. Compare duplicated records.
4. Determine whether they represent genuine repeated observations.
5. Remove only confirmed duplicate records.

6. Record the number of removed records.

Example key fields could include:

Crop + Location + Year + Market

However, the exact duplicate key should depend on the dataset structure.

16. Handling Inconsistent Data

Agricultural datasets may contain inconsistent naming and formatting.

Examples:

- Maharashtra
- maharashtra
- MH

These may represent the same state.

Similarly:

- Wheat
- wheat
- WHEAT

may represent the same crop.

Proposed solution

Create standard formats for:

- State names
- District names
- Crop names
- Units
- Dates
- Categories

A standardized reference table can be used where necessary.

17. Data Type Validation

Each variable should have an appropriate data type.

Examples:

Variable	Expected Type
Crop Name	Text
State	Text
Production	Numeric
Yield	Numeric
Price	Numeric
Year	Integer
Date	Date

Incorrect data types should be identified and converted before analysis.

18. Outlier Detection

Outliers are observations that differ substantially from the general pattern of the data.

Potential causes include:

- Data-entry errors
- Measurement errors
- Incorrect units
- Genuine unusual events

Detection Methods

Possible techniques include:

- Box plots
- Interquartile Range (IQR)
- Z-score
- Distribution analysis

Important Principle

An outlier should not automatically be deleted.

The observation should first be investigated to determine whether it is:

1. An error,
2. A valid but unusual observation, or
3. A result of a different measurement condition.

Only confirmed erroneous values should be corrected or removed.

19. Range and Logical Validation

Values should be checked against reasonable constraints.

Examples:

- Production should not be negative.
- Cultivated area should not be negative.
- Prices should not contain impossible negative values unless explicitly defined by the source.
- Year values should fall within the expected study period.
- Crop names should belong to the selected crop categories.

These checks help identify errors that basic missing-value or duplicate checks may not detect.

20. Manual and Automated Cleaning Approaches

Manual Approach

Manual cleaning can be performed using tools such as Microsoft Excel.

Useful operations include:

- Filtering
- Sorting
- Removing duplicates
- Finding blanks
- Standardizing categories
- Checking unusual values
- Creating validation rules

Advantages

- Easy for small datasets
- User-friendly
- Useful for reviewing individual records

Limitations

- Time-consuming for large datasets
 - Higher risk of human error
 - Difficult to reproduce consistently
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21. Automated Cleaning Approach

Python can be used for repeatable data-cleaning workflows.

Potential libraries include:

- Pandas
- NumPy

- Matplotlib
- Seaborn

A typical automated workflow can include:

Load Data → Inspect → Clean → Validate → Save Clean Dataset

Automation is especially useful when datasets are large or need to be processed repeatedly.

22. Proposed Tools and Technologies

Tool	Purpose
Microsoft Excel	Manual inspection, cleaning and basic analysis
Python	Automated data processing
Pandas	Data loading and manipulation
NumPy	Numerical operations
Matplotlib	Visualization
Seaborn	Statistical visualization
Jupyter Notebook	Interactive analysis and documentation
CSV/Excel	Data storage and exchange
API tools	Automated data collection where available

23. Data Quality Validation Framework

After cleaning, the dataset should be validated before analysis.

The following dimensions should be checked:

Accuracy

Does the data represent the intended information correctly?

Completeness

Are important fields sufficiently populated?

Consistency

Are formats, units, and categories consistent?

Uniqueness

Are duplicate records removed appropriately?

Validity

Do values follow expected formats and logical rules?

Timeliness

Is the data appropriate for the required analysis period?

24. Proposed Data Quality Checklist

Before using the dataset for analysis, the following checklist should be completed:

- Source documented
- Dataset download/retrieval date recorded
- Variables documented
- Data types checked
- Missing values analyzed
- Duplicates checked
- Categories standardized
- Units standardized
- Outliers investigated
- Invalid values checked

- Date fields validated
 - Final dataset reviewed
 - Original dataset preserved
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25. Proposed End-to-End Workflow

The complete proposed workflow is:

Business Question



Identify Required Data



Research Public Sources



Evaluate Data Sources



Collect Data



Preserve Original Dataset



Initial Data Inspection



Handle Missing Values



Remove Confirmed Duplicates



Standardize Formats and Categories



Detect and Investigate Outliers



Perform Logical Validation



Quality Check



Clean Dataset Ready for EDA

26. Data Documentation and Reproducibility

To make the analysis reproducible, important information should be recorded during data collection and cleaning.

The documentation should include:

- Dataset name
- Source organization
- Source URL
- Download/retrieval date
- File format
- Variables used
- Data-cleaning operations
- Missing-value treatment
- Duplicate treatment
- Outlier treatment
- Assumptions
- Limitations

The original dataset should be preserved separately from the cleaned dataset so that changes can be traced.

27. Potential Risks and Mitigation Strategies

Risk	Mitigation
Unreliable source	Prefer official/institutional sources
Missing data	Analyze missingness before treatment
Duplicate records	Apply duplicate checks
Inconsistent units	Standardize units
Incorrect values	Apply range and logical checks
Scraping failures	Prefer APIs/downloads where available
Changing data sources	Record dataset versions and dates
Over-cleaning	Preserve original data
Loss of information	Document every major transformation

28. Expected Outcomes

The proposed strategy is expected to provide:

- A structured approach to agribusiness data collection.
- Identification of suitable public data sources.
- A repeatable data-cleaning process.
- Improved data consistency and quality.
- Documentation of data-processing decisions.
- A reliable foundation for exploratory analysis.
- Better reproducibility of future analysis.

The cleaned dataset developed through this framework can subsequently be used for exploratory data analysis, statistical analysis, market analysis, and forecasting.

29. Limitations

This Week 1 report is a data collection and cleaning strategy rather than a completed analytical study.

Therefore:

- No final dataset is claimed to have been collected through this report.
- No actual statistical results are presented.
- No agricultural trend is claimed as a finding.
- The proposed methods may need modification depending on the selected dataset.
- Data availability and quality may vary between sources.
- Some sources may have access restrictions or different licensing conditions.

These limitations will be addressed during subsequent stages if actual data is collected.

30. Conclusion

A reliable data-analysis project begins with reliable data collection and systematic cleaning. In the agribusiness sector, data can come from multiple sources and may contain missing values, duplicates, inconsistent formats, outliers, and other quality issues.

This report proposed a structured framework for identifying public agribusiness data sources, selecting appropriate collection methods, documenting data, and preparing the data for analysis.

The strategy combines manual tools such as Excel with automated Python-based methods using libraries such as Pandas and NumPy. Data-quality checks are proposed to improve accuracy, completeness, consistency, validity, and reproducibility.

The framework will provide a foundation for the next stages of the internship, particularly exploratory data analysis, visualization, statistical analysis, market analysis, and forecasting.

31. References

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